JAY SHAH
AI/ML Engineer | Generative AI, RAG & Agentic Systems
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PROFESSIONAL SUMMARY:

AI/ML Engineer with 5 years of experience building and deploying machine learning and generative AI systems in healthcare, banking and insurance. Specializes in production RAG pipelines, LLM fine-tuning (LoRA/QLoRA), and multi-agent workflows using LangChain, LangGraph, and CrewAI, backed by MLOps/LLMOps on AWS, Azure, and GCP. Builds the full path from data pipelines and model development to containerized APIs, CI/CD, monitoring, and HIPAA-aware security. Currently delivering RAG and agentic AI over 2M+ patient records and 500K+ research documents at Komodo Health.

CORE TECHNICAL SKILLS:

Languages: Python, SQL, PySpark.
Generative AI & LLMs: RAG, Agentic AI/Multi-Agent Systems, LangChain, LangGraph, CrewAI, AutoGen, LlamaIndex, OpenAI, Claude, Gemini, LLaMA2, Hugging Face, Prompt Engineering, LoRA/QLoRA (PEFT), RLHF/DPO, Embeddings, Semantic Search.
Vector Databases: FAISS, Pinecone, Weaviate, Milvus, PostgreSQL vector store.
LLM Evaluation & Observability: RAGAS, DeepEval, LangSmith, Hallucination Detection.
Machine & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, Transformers (BERT, T5), CNNs, LSTM, Time-Series Forecasting, Anomaly Detection, NLP, Computer Vision (YOLO, U-Net, Mask R-CNN, OpenCV).
MLOps & Deployment: MLflow, Kubeflow, Airflow, Docker, Kubernetes, FastAPI, Flask, Terraform, CI/CD (Jenkins), SageMaker Pipelines, Model Monitoring, Drift Detection, Prometheus, Grafana.
Cloud & Data Platforms: AWS (SageMaker, Lambda, S3, Glue), Azure (Azure ML, AKS, Functions, Data Factory, Synapse, Key Vault), GCP (Vertex AI, BigQuery, Dataflow), Databricks, Snowflake, Spark, Kafka.
Databases: PostgreSQL, MySQL, MongoDB, Snowflake, Neo4j.
Security & Compliance: HIPAA, IAM, RBAC, OAuth2, Encryption, Responsible AI, Explainable AI (XAI).
Analytics: Statistical Modeling, A/B Testing, Hypothesis Testing, Feature Engineering, Pandas, NumPy, Power BI, Tableau.

PROFESSIONAL EXPERIENCE:

Komodo Health — Remote (New York, NY) **May 2025 – Present**
Senior AI/ML Engineer – Generative AI & Agentic Systems

- Built enterprise RAG and agentic AI systems for healthcare automation using OpenAI, Claude, Gemini, LangChain, LangGraph, CrewAI, and AutoGen.
- Developed RAG pipelines on FAISS, Pinecone, Weaviate, and Milvus over 2M+ patient records and 500K+ research documents, reducing hallucination rate to below 3%.
- Fine-tuned LLaMA2 (70B) using LoRA/QLoRA, RLHF, and DPO, cutting inference cost by 60% while improving task performance.
- Built NLP and GenAI applications (document summarization, semantic search, intelligent assistants) using BERT, T5, Hugging Face, spaCy, and LlamaIndex.
- Implemented MLOps and LLMOps pipelines with MLflow, Kubeflow, Airflow, Docker, Kubernetes, Terraform, and CI/CD, covering training, deployment, and monitoring.
- Engineered real-time and batch data pipelines with Spark, Kafka, Databricks, Snowflake, AWS Glue, and Google Dataflow for large-scale ETL and streaming analytics.
- Developed computer vision and medical imaging models (YOLOv11, U-Net, Mask R-CNN, ResNet50) in PyTorch for surgical analytics and diagnostic imaging.
- Built predictive and anomaly detection models (LSTM, XGBoost, Random Forest, Prophet, ARIMA) for sepsis prediction and healthcare forecasting.
- Deployed HIPAA-compliant AI services with IAM, OAuth2, RBAC, and encryption; partnered with data scientists, product managers, and clinical stakeholders.

Environment: Python, SQL, OpenAI, Claude, Gemini, LangChain, LangGraph, CrewAI, FAISS, Pinecone, PyTorch, TensorFlow, Hugging Face, Spark, Kafka, Databricks, Snowflake, Airflow, MLflow, Docker, Kubernetes, AWS, Azure ML, GCP, PostgreSQL.

Jupiter (Banking) — Remote **January 2023 – July 2024**
Data Scientist – Generative AI & Machine Learning

- Built RAG-based LLM applications and enterprise chatbots using OpenAI GPT-3.5, LLaMA2, LangChain, and embeddings stored in a PostgreSQL vector database for knowledge retrieval and workflow automation.
- Designed a configurable chatbot framework in Streamlit with prompt-engineering patterns to improve customer engagement and operational efficiency.
- Fine-tuned BERT and domain-specific NLP models to improve semantic understanding, intent classification, and response accuracy.
- Trained and deployed ML/DL models (Scikit-learn, TensorFlow, Keras, PyTorch) for predictive analytics, text analytics, and customer intelligence on large-scale datasets.
- Built distributed training and ETL pipelines using Azure Databricks, PySpark, MLflow, AWS SageMaker, AWS Glue, Azure Data Factory, and Synapse.
- Served models as REST APIs and serverless endpoints using FastAPI, Azure Kubernetes Service, Azure Functions, and SageMaker endpoints.
- Automated CI/CD and model monitoring with Jenkins, Bitbucket, SageMaker Pipelines, MLflow, and Model Monitor, reducing deployment time and improving model lifecycle management.
- Secured ML deployments with Azure Key Vault, IAM, and RBAC; monitored API and middleware performance using APIGEE and Splunk.

Environment: Python, Scikit-learn, TensorFlow, PyTorch, Hugging Face, LangChain, OpenAI, FastAPI, AWS (Glue, SageMaker, Lambda, RDS), Azure (Databricks, ADF, Synapse, AKS, ML Studio), Jenkins, Bitbucket, Splunk.

Generali Central Life Insurance — India February 2021 – December 2022

Data Scientist – Machine Learning

- Built and deployed ML models for forecasting, credit risk, customer churn, classification, and anomaly detection across batch and streaming environments.
- Developed scalable ETL and feature pipelines with Spark, PySpark, Hadoop, Hive, and Kafka for predictive analytics and real-time data processing.
- Used GCP (BigQuery, Vertex AI, Dataflow) and TensorFlow for distributed training, deployment, and large-scale analytics.
- Implemented MLOps workflows with Databricks, MLflow, Azure ML, and AWS SageMaker for CI/CD, validation, drift detection, and retraining.
- Delivered NLP chatbots, recommendation systems, and CNN-based computer vision applications; built Tableau and Power BI dashboards for stakeholders.

Environment: Python, Scikit-learn, TensorFlow, PySpark, Hadoop, Hive, Kafka, AWS, GCP, Azure, Databricks, MLflow, Neo4j, Tableau, Power BI, SQL.

EDUCATION:

- Master of Science in Computer Science from The University of Texas, TX, USA.
- Bachelors from Charusat University, India.